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School of Computer Science and Artificial Intelligence

Lab Assignment 14.5

Program	**: B. Tech (CSE)**
Specialization	**: -**
Course Title	**: AI Assisted Coding**
Course Code	**: 23CS002PC304**
Semester	**:II**
Academic Session	**: 2025-2026**
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Enrollment No.	**: 2403A51L53**
Batch No :	**: 52**
Date	**: 13/03/26**

Task 1 – Image Gallery Website

Prompt:

Create an image gallery web page.

Requirements:

• Display images in a grid layout.

• Use AI to:

- Optimize layout using CSS Grid.
- Add hover zoom or overlay effects.
- Ensure mobile responsiveness.

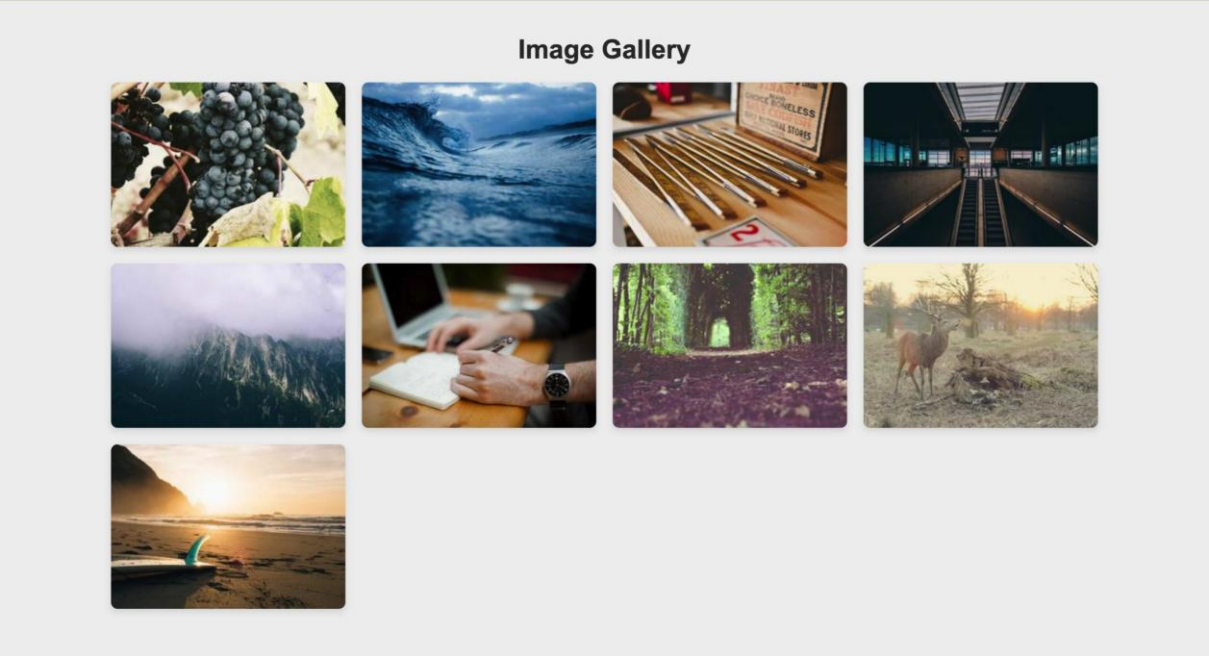
Code:

index.html > ...

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Image Gallery</title>
    <link rel="stylesheet" href="styles.css">
</head>
<body>
    <h1>Image Gallery</h1>
    <div class="gallery">
        <div class="gallery-item">
            <img src="https://picsum.photos/300/200?random=1" alt="Random Image 1">
            <div class="overlay">
                <p>Image 1</p>
            </div>
        </div>
        <div class="gallery-item">
            <img src="https://picsum.photos/300/200?random=2" alt="Random Image 2">
            <div class="overlay">
                <p>Image 2</p>
            </div>
        </div>
        <div class="gallery-item">
            <img src="https://picsum.photos/300/200?random=3" alt="Random Image 3">
            <div class="overlay">
                <p>Image 3</p>
            </div>
        </div>
        <div class="gallery-item">
            <img src="https://picsum.photos/300/200?random=4" alt="Random Image 4">
            <div class="overlay">
                <p>Image 4</p>
            </div>
        </div>
        <div class="gallery-item">
            <img src="https://picsum.photos/300/200?random=5" alt="Random Image 5">
            <div class="overlay">
                <p>Image 5</p>
            </div>
        </div>
        <div class="gallery-item">
            <img src="https://picsum.photos/300/200?random=6" alt="Random Image 6">
            <div class="overlay">
                <p>Image 6</p>
```

```html
            </div>
        </div>
        <div class="gallery-item">
            <img src="https://picsum.photos/300/200?random=7" alt="Random Image 7">
            <div class="overlay">
                <p>Image 7</p>
            </div>
        </div>
        <div class="gallery-item">
            <img src="https://picsum.photos/300/200?random=8" alt="Random Image 8">
            <div class="overlay">
                <p>Image 8</p>
            </div>
        </div>
        <div class="gallery-item">
            <img src="https://picsum.photos/300/200?random=9" alt="Random Image 9">
            <div class="overlay">
                <p>Image 9</p>
            </div>
        </div>
    </div>
</body>
</html>
```

Output:



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Task 2 – Profile Card Generator

Prompt: Create a profile card web page.

Requirements:

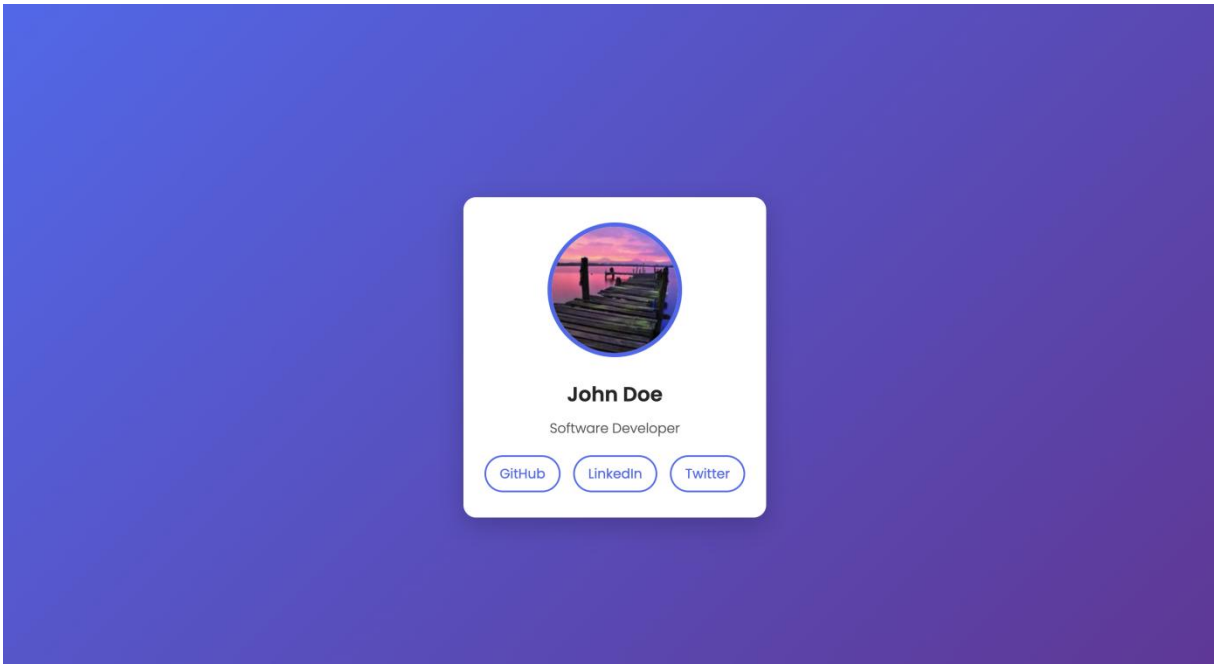
• Display user image, name, role, and social media links.
• Use AI to:
- Suggest modern UI colors and fonts.
- Design the card using Flexbox.
- Add hover animation and shadow effects

Code:

profile.html > html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Profile Card</title>
    <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&display=
    <link rel="stylesheet" href="profile.css">
</head>
<body>
    <div class="profile-card">
        <div class="profile-image">
            <img src="https://picsum.photos/150/150?random=10" alt="Profile Picture">
        </div>
        <div class="profile-info">
            <h1 class="name">Kowshik</h1>
            <p class="role">Software Developer</p>
            <div class="social-links">
                <a href="#" class="social-link" data-platform="github">GitHub</a>
                <a href="#" class="social-link" data-platform="linkedin">LinkedIn</a>
                <a href="#" class="social-link" data-platform="twitter">Twitter</a>
            </div>
        </div>
    </div>
</body>
</html>
```

Output:



Task 3 -Develop a dynamic shopping cart system.

Prompt: Create a shopping cart system.

Requirements:

- Add, remove, and update product quantities.
- Display real-time total price.
- Use AI to:
  - Manage cart state using JavaScript objects.
  - Persist cart data in localStorage

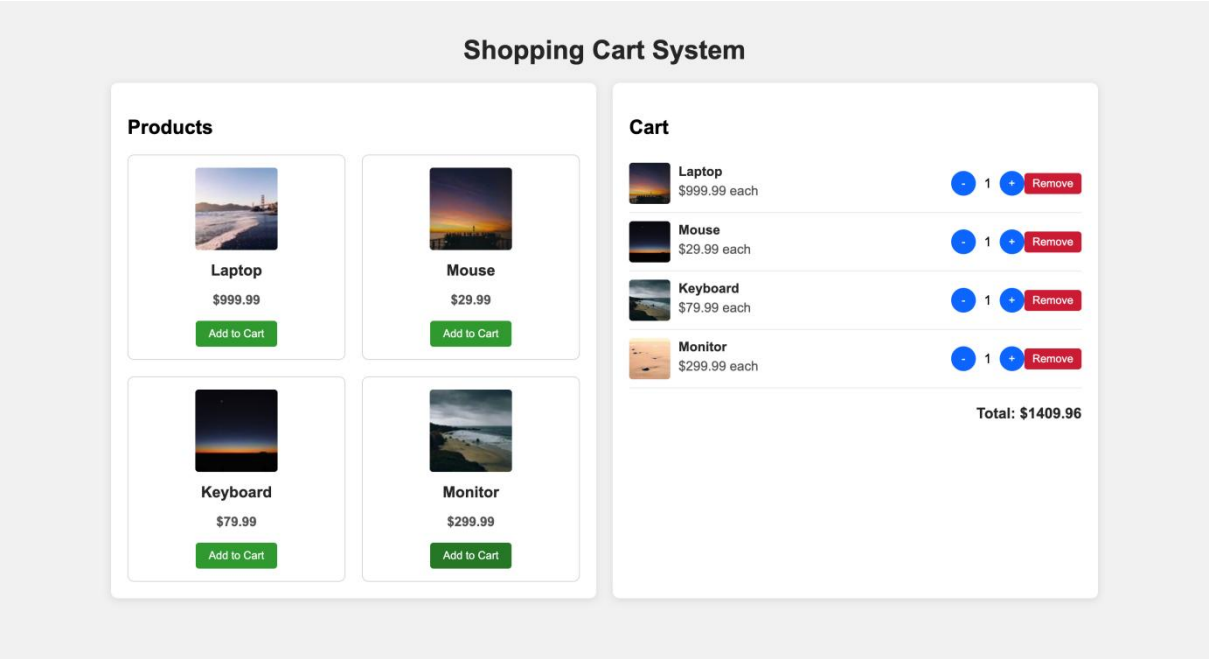
Code:

cart.html > ...

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Shopping Cart</title>
    <link rel="stylesheet" href="cart.css">
</head>
<body>
    <h1>Shopping Cart System</h1>
    <div class="container">
        <div class="products">
            <h2>Products</h2>
            <div class="product-list">
                <div class="product" data-id="1">
                    <img src="https://picsum.photos/100/100?random=20" alt="Product 1">
                    <h3>Laptop</h3>
                    <p>$999.99</p>
                    <button class="add-to-cart">Add to Cart</button>
                </div>
                <div class="product" data-id="2">
                    <img src="https://picsum.photos/100/100?random=21" alt="Product 2">
                    <h3>Mouse</h3>
                    <p>$29.99</p>
                    <button class="add-to-cart">Add to Cart</button>
                </div>
                <div class="product" data-id="3">
                    <img src="https://picsum.photos/100/100?random=22" alt="Product 3">
                    <h3>Keyboard</h3>
                    <p>$79.99</p>
                    <button class="add-to-cart">Add to Cart</button>
                </div>
                <div class="product" data-id="4">
                    <img src="https://picsum.photos/100/100?random=23" alt="Product 4">
                    <h3>Monitor</h3>
                    <p>$299.99</p>
                    <button class="add-to-cart">Add to Cart</button>
                </div>
            </div>
        </div>
        <div class="cart">
            <h2>Cart</h2>
            <div class="cart-items"></div>
            <div class="cart-total">
```

```html
                <strong>Total: $<span id="total">0.00</span></strong>
            </div>
        </div>
    </div>
    <script src="cart.js"></script>
</body>
</html>
```

Output:



Task 4 – Role-Based Access Control Application

Prompt: Create a role-based web application.

Requirements:

- Admin, Editor, User roles.
- Conditional UI rendering.
- Use AI to:
    - Design access control logic.
    - Maintain secure state transitions.
    - Create scalable UI components.

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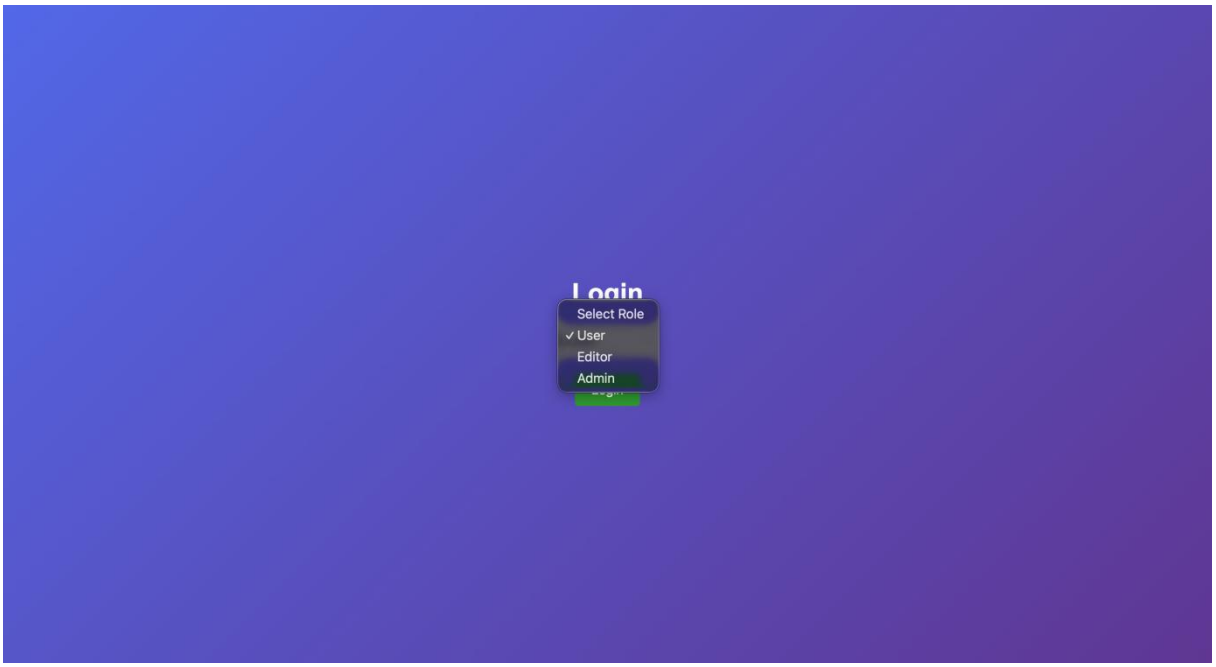
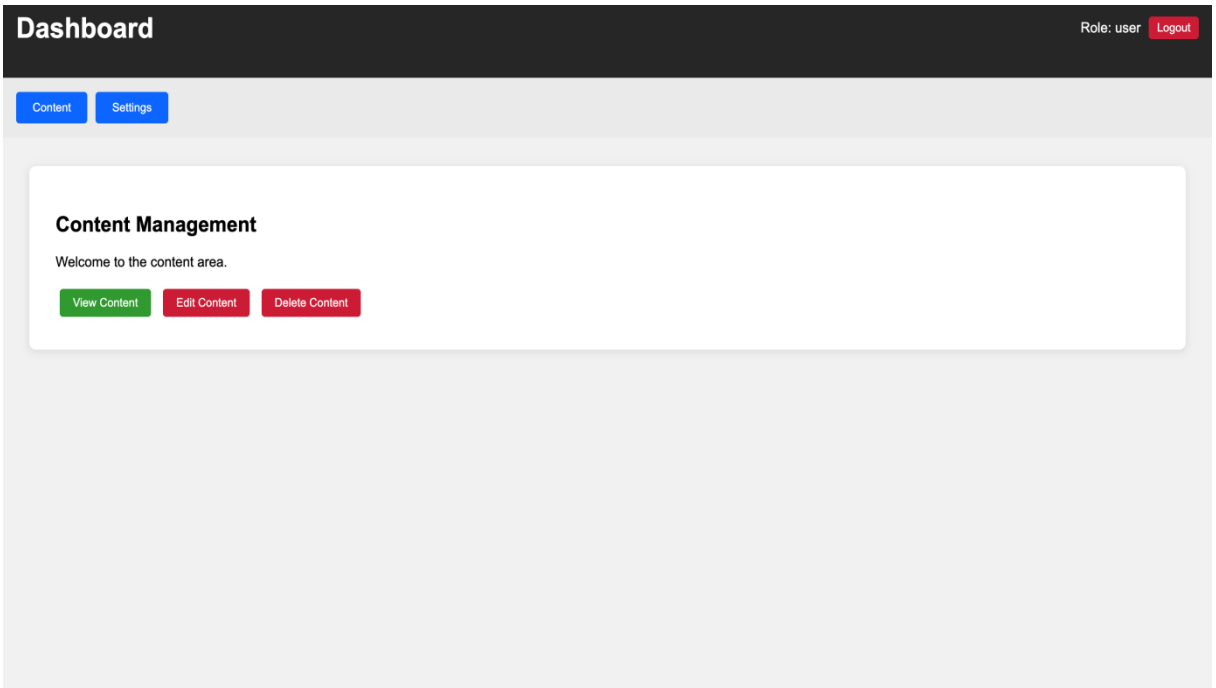
Code:

```html
rbac.html > ...
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Role-Based Access Control</title>
    <link rel="stylesheet" href="rbac.css">
</head>
<body>
    <div id="login" class="login-container">
        <h1>Login</h1>
        <select id="role-select">
            <option value="">Select Role</option>
            <option value="user">User</option>
            <option value="editor">Editor</option>
            <option value="admin">Admin</option>
        </select>
        <button id="login-btn">Login</button>
    </div>

    <div id="dashboard" class="dashboard" style="display: none;">
        <header class="header">
            <h1>Dashboard</h1>
            <div class="user-info">
                <span id="current-role">Role: </span>
                <button id="logout-btn">Logout</button>
            </div>
        </header>

        <nav class="nav">
            <button class="nav-btn" data-section="content">Content</button>
            <button class="nav-btn" data-section="users" id="users-nav">Users</button>
            <button class="nav-btn" data-section="settings" id="settings-nav">Settings</b
        </nav>

        <main class="main">
            <section id="content-section" class="section">
                <h2>Content Management</h2>
                <p>Welcome to the content area.</p>
                <button id="view-content-btn" class="action-btn">View Content</button>
                <button id="edit-content-btn" class="action-btn">Edit Content</button>
                <button id="delete-content-btn" class="action-btn">Delete Content</button
            </section>
```

```html
            <section id="users-section" class="section" style="display: none;">
                <h2>User Management</h2>
                <p>Manage users here.</p>
                <button id="view-users-btn" class="action-btn">View Users</button>
                <button id="add-user-btn" class="action-btn">Add User</button>
                <button id="delete-user-btn" class="action-btn">Delete User</button>
            </section>

            <section id="settings-section" class="section" style="display: none;">
                <h2>Settings</h2>
                <p>Application settings.</p>
                <button id="view-settings-btn" class="action-btn">View Settings</butto
                <button id="edit-settings-btn" class="action-btn">Edit Settings</butto
            </section>
        </main>
    </div>

    <script src="rbac.js"></script>
</body>
</html>
```

Output:


Login
Select Role
User
Editor
Admin


Dashboard
Role: user
Logout
Content
Settings
Content Management
Welcome to the content area.
View Content
Edit Content
Delete Content

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Task 5 – Multi-Language Website

Prompt:Design a multilingual website.

Requirements:

1. Replace unclear variable names with meaningful descriptive names.
2. Add inline comments explaining the logic.
3. Ensure the program output remains exactly the same.
4. Follow Python naming conventions.
5. Make the code easier to understand for beginners.

Original Code:

```
x = 50
y = 30
z = x + y
print(z)
```

Output:

```
first_number = 50
second_number = 30

# Add the two numbers together
sum_of_numbers = first_number + second_number

# Print the result (same output as before)
print(sum_of_numbers)
Ctrl+L to chat, Ctrl+K to generate
```